

HENRIQUE G. LEITÃO

henrique.grancha.leitao@gmail.com
ORCID | Google Scholar | LinkedIn

PROFILE

Evolutionary genomics and bioinformatics researcher completing a PhD at the University of Antwerp. My work combines chromosome-level genome assembly, population genomics, phylogenomics, genotype-phenotype analysis and RNA-seq, with a focus on osteoderm variation in the Cape cliff lizard (*Hemicordylus capensis*). Male-female whole-genome resequencing identified a putative Y chromosome and provided the first evidence for an XX/XY sex-determination system in Cordylidae. Building on this work, I aim to study comparative sex-chromosome evolution, using chromosome-resolved genomes and natural populations to understand why sex chromosomes remain homomorphic, differentiate or turn over.

EDUCATION

- PhD candidate in Evolutionary Genomics** - University of Antwerp, Belgium 2021-present
- FWO-funded fellowship; thesis: "The genomic basis of rapid change in a functionally significant trait: osteoderm evolution in a girdled lizard"; submission targeted for autumn 2026.
- MSci in Zoology, First Class Honours** - University of Glasgow, United Kingdom 2020
- Ranked 6th of 75 students in the graduating cohort (top 8%); final honours GPA: 18.9/22.
 - Honours project: RNA extraction, RNA-seq differential-expression analysis and electron microscopy on ventral colour variation in common lizards, with results published in *Molecular Ecology* (2024).
- International Asian Studies Programme, Zoology/Animal Biology** - The Chinese University of Hong Kong 2016-2017

RESEARCH EXPERIENCE

- FWO PhD Fellow** - University of Antwerp 2021-present
- Built the *H. capensis* chromosome-level reference genome end-to-end; led laboratory workflows, final QC, public release and accession submission for three ERGA Pilot Project genomes using PacBio HiFi, Oxford Nanopore, Illumina and Hi-C data.
 - Conducted population-genomic, phylogenomic and demographic analyses, including GWAS of osteoderm phenotypes in 241 individuals across seven populations and RNA-seq differential-expression analysis.
 - Used male-female whole-genome resequencing, coverage and heterozygosity patterns to identify the putative Y chromosome of *H. capensis* and provide the first evidence for an XX/XY sex-determination system in Cordylidae.
 - Built and documented reproducible Python, R and Bash/Unix workflows for variant calling, population structure, differential expression and comparative genomics, and ran them on Slurm-based HPC systems.
- Research Assistant** - Uppsala University, Sweden 2018-2019
- Full-year computational and laboratory placement involving ancient-DNA extraction, NGS library preparation, mapping, contamination assessment, metagenomics and host/dietary genomic analysis; results published in *Molecular Biology and Evolution* (2020) and *Current Biology* (2021).

RESEARCH LEADERSHIP

- Hi-C Sequencing Hub Lead and Genome Team Member** - European Reference Genome Atlas (ERGA) 2021-2023
- Led the Antwerp Hi-C sequencing hub, managing sample intake and storage, database records, manifests and metadata, Hi-C/Omni-C and Illumina library preparation, sequencing QC and data delivery across Europe.
 - Coordinated sample logistics, communication and data generation across 17 genome projects, contributing to laboratory workflows, quality control and delivery of genome assembly resources.

FIELDWORK AND ORGANISMAL EXPERIENCE

- Field Assistant** - University of Glasgow, Carinthia, Austria 2016-2018
- Contributed to a study on the evolutionary history of viviparity in lizards during three field seasons (seven months total).
 - Conducted lizard capture, behavioural experiments, morphometric data collection and embryonic sampling; assisted in husbandry and spectrophotometry measurements for pigmentation studies.
- Field Assistant** - The Chinese University of Hong Kong, Hong Kong 2017
- Participated in research on genetic diversity of atyid shrimp across Hong Kong; collected water and shrimp samples, measured flow rates and pH, and recorded spatial data using GPS.

TECHNICAL SKILLS

Programming	Python, R, Bash/Unix shell, Git/GitHub
Software and formats	FASTA/FASTQ, BAM/VCF/GFF, fastp, BWA-MEM, SAMtools/BCFtools, PLINK, GEMMA, STAR, DESeq2, BUSCO, Merqury, BlobToolKit, Kraken2
Genome resources	Four chromosome-level reference-genome projects with public release, accessioning and assembly QC, using PacBio HiFi, ONT, Illumina/BGI short-read and Hi-C/Omni-C data
Genomic analysis	Population structure, PCA, phylogenomics, demographic inference, GWAS, genotype-phenotype analysis, sex-linked sequence analysis, RNA-seq and differential expression
Wet-lab methods	High-molecular-weight and standard DNA extraction, ancient DNA, Illumina, Hi-C/Omni-C and RNA-library preparation, RNA extraction
Computing	HPC, Slurm, Snakemake, Conda, reproducible workflows, quality control and data provenance

PUBLICATIONS

Full list: [Google Scholar](#)

1. Drewalowski J, Kliver S, Hilgers L, Møller PR, Mak SST, Kovačić I, Petersen B, Nesme J, McCartney AM, Mouton A, Formenti G, Svartal H, Diedericks G, **Leitão HG** et al. (2026; preprint posted 3 September). Sea stickleback genome reveals repeated chromosomal rearrangements in sticklebacks. *bioRxiv*. [DOI](#)
2. Potente G, Yousefi N, Choudhury RR, Grob S, Gavriliu IA, Keller B, Mora-Carrera EM, Szövényi P, Stubbs RL, Weiss-Schneeweiss H, Temsch EM, Schneeweiss GM, Hoffmann MH, Formenti G, McCartney AM, Mouton A, **Leitão HG** et al. (2026; published 10 June). Distinct genomic architectures but the same gene underlie the convergent evolution of a plant supergene. *Science Advances*, 12(24), eaec1996. [DOI](#)
3. **Leitão HG**, Bantounas A, Diedericks G, Broeckhoven C, Van Damme R, Matthysen E, Svartal H. (2026; published 2 June). Phylogenomic and demographic history of the Cape cliff lizard (*Hemicordylus capensis*). *Journal of Evolutionary Biology*, 39(7), 891-899. [DOI](#)
4. Rödin-Mörch P, Bunikis I, Choi E, Ciofi C, Diedericks G, Diroma MA, Einarsdóttir E, Forsäter K, Heintz J, Jonsäll L, Lantz H, Laurila A, **Leitão HG** et al. (2025). A chromosome-level genome assembly of the European green toad (*Bufo viridis*). *G3: Genes, Genomes, Genetics*, 15(3), jkaf002. [DOI](#)
5. Kurta K, Fedi MO, Baker K, Barker T, Catchpole L, Ciofi C, Cocco A, Collins J, Diedericks G, Diroma MA, Durrant A, Hindar K, Iannucci A, Irish N, Knitthoffer V, Laikre L, **Leitão HG** et al. (2025). Whole genome sequencing reveals how plasticity and genetic differentiation underlie sympatric morphs of Arctic charr. *Molecular Ecology*, 34(19), e70085. [DOI](#)
6. Massa P, **Leitão HG**, Manousaki T, Cariani A, Svartal H, Tsaparis D et al. (2025). Genomic insights into the population structure and adaptive variation of *Mullus barbatus* in the Mediterranean Sea. *BMC Ecology and Evolution*, 25, 101. [DOI](#)
7. McCartney AM, Formenti G, Mouton A, De Panis D, Marins LS, **Leitão HG** et al. (2024). The European Reference Genome Atlas: piloting a decentralised approach to equitable biodiversity genomics. *npj Biodiversity*, 3(1), 28. [DOI](#)
8. Marques JP, Alves PC, Amorim IR, Lopes RJ, Moura M, Myers EW, **Leitão HG** et al. (2024). Building a Portuguese coalition for biodiversity genomics. *npj Biodiversity*, 3(1), 27. [DOI](#)
9. Recknagel H, **Leitão HG**, Elmer KR. (2024). Genetic basis and expression of ventral colour in polymorphic common lizards. *Molecular Ecology*, 33(5), e17278. [DOI](#)
10. **Leitão HG**, Diedericks G, Broeckhoven C, Baeckens S, Svartal H. (2023). Chromosome-level genome assembly of the Cape cliff lizard (*Hemicordylus capensis*). *Genome Biology and Evolution*, 15(2), evad001. [DOI](#)
11. Brealey JC, **Leitão HG**, Hofstede T, Kalthoff DC, Guschanski K. (2021). The oral microbiota of wild bears in Sweden reflects the history of antibiotic use by humans. *Current Biology*, 31(20), 4650-4658.e6. [DOI](#)
12. Brealey JC, **Leitão HG**, van der Valk T, Xu W, Bougiouri K, Dalén L, Guschanski K. (2020). Dental calculus as a tool to study the evolution of the mammalian oral microbiome. *Molecular Biology and Evolution*, 37(10), 3003-3022. [DOI](#)
13. Recknagel H, Layton M, Carey R, **Leitão H**, Sutherland M, Elmer KR. (2018). Melanism in common lizards (Squamata: Lacertidae: *Zootoca vivipara*): new evidence for a rare but widespread ancestral polymorphism. *Herpetology Notes*, 11, 607-612.

MANUSCRIPTS IN PREPARATION

1. **Leitão HG** et al. (in preparation). Genomic and transcriptomic insights into osteoderm variation in the Cape cliff lizard (*Hemicordylus capensis*).
2. **Leitão HG** et al. (in preparation). Genome-wide phylogenomics establishes a family-wide evolutionary framework for Cordylidae.

PRESENTATIONS

- **Leitão HG** (2022). Introduction to the ERGA pilot project. **Biodiversity Genomics 2022**, online (talk).
- **Leitão HG**, Diedericks G, Broeckhoven C, Baeckens S, Van Damme R, Svartal H (2022). Chromosome-level assembly of the Cape cliff lizard. **Biodiversity Genomics 2022**, online (poster and lightning talk).
- **Leitão HG**, Recknagel H, Elmer KR (2021). Genetic mechanisms of colour divergence in common lizards. **Evolution 2021**, online (talk).
- **Leitão HG**, Brealey JC, Guschanski K (2019). Recovering an authentic oral microbiome signature and host genetic diversity from historical dental calculus of non-human mammals. **SMBE 2019**, Manchester (poster).
- **Leitão HG**, Brealey JC, Guschanski K (2019). How to identify 'true' oral taxa in dental calculus? **Revolution 2019**, Animal Ecology, Uppsala University internal symposium, Uppsala, Sweden (poster).

FELLOWSHIPS AND AWARDS

FWO PhD Fellowship	2021
Undergraduate Travel and Mentoring Award, SMBE19	2019
Erasmus+ Traineeship Grant	2018
Dr Victor Lim Travel Scholarship	2016

TEACHING

Guest lecturer in Herpetology - University of Antwerp	2022, 2024
Teaching assistant - Omics Techniques, Laboratory Skills and Island Biology, University of Antwerp	2022-2023